

Lab 2: Data Visualization & Summary Statistics

STAT 141, Week 2

Your name here

In this lab, you will practice

1. Creating and interpreting `ggplot2` graphs.
2. Producing and interpreting summary statistics.

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Remember that you are not allowed to use any internet resources or generative AI models (like ChatGPT or Claude) on your assignment. While there are many ways to achieve tasks in R, you may only use packages and functions that have been covered in this course. If you are unsure of how to approach an exercise or of what function to use, help is available to you! Please visit office hours, collaborate with your peers, or post in the Slack channel.

```
# Load the necessary packages
library(lubridate)
library(tidyverse) # Includes ggplot2
```

Exercises

Exercise 1: Visualizing somerville with barplots

For Problems 1 and 2, we will use the `somerville` dataset that you started exploring in Homework 1. This version of the dataset in the `somerville_clean.csv` file has been cleaned, with the 999 values replaced with NA. The atypically large values for `age` have also been converted to NA.

Starting in 2011, Somerville city officials decided to start measuring residents' overall happiness and satisfaction with many aspects of life in Somerville, MA; their goal was to become the first city in the United States to systematically track people's happiness. The Happiness Survey is an initiative of the Mayor's Office of Analytics and Innovation (SomerStat) and has the goal of supporting the local government's understanding of what makes Somerville a great place to live.

The happiness survey is sent out every two years to a random sample of Somerville households, with the instruction that each survey should be filled out by one person. The survey asks residents to rate their personal happiness, wellbeing, and satisfaction with City services. This combined dataset includes the survey responses from 2011 to 2019.

Run the code in the following chunk to load the cleaned version of the `somerville` data.

```
somerville <- read.csv("data/somerville_clean.csv")
```

Variables in the `somerville` data:

- `year`: year that response was collected
- `happy`: happiness rating, on a scale from 1 to 10 with 10 being most happy
- `satisfied_life`: rating for satisfaction with life in general, on a scale from 1 to 10 with 10 being most satisfied
- `satisfied_somerville`: rating for satisfaction with Somerville as a place to live, on a scale from 1 to 10 with 10 being most satisfied
- `age`: age in years
- `children_home`: whether respondent has children under 18 living at home
- `ward`: neighborhood ward the respondent lives in
- `precinct`: neighborhood precinct the respondent lives in
- `student`: whether respondent is a student
- `convenient_transport`: rating for how convenient it is to get to where you want to go, on a scale from 1 to 10 with 10 being most satisfied
- `direction_somerville`: whether respondent feels the City is headed in the right direction or is on the wrong track
- `housing`: whether respondent rents or owns their home
- `primary_transport`: primary form of transportation
- `lived_here`: years respondent has lived in Somerville

a. Let's start by getting a sense of how people get around Somerville. Fill in the blanks (---) to create a **bar graph** of the variable `primary_transport`. Make sure to put the completed code into an R chunk. [Only code is needed for this question]

```
ggplot(data = ---, mapping = aes(x = ---)) +  
  geom_---()
```

```
# Create bar graph
```

b. Create two bar plots of the variables `primary_transport` and `direction_somerville`:
+ For the first one, display the **counts** of each category of `direction_somerville` within `primary_transport` groups (bars). + For the second one, display the **proportions** of each category of `direction_somerville` within `primary_transport` groups (bars).

Compare and contrast the information provided in the two plots. [Code and text are needed for this question]

```
# Bar plot: Counts
```

```
# Bar plot: Proportions
```

c. Draw two conclusions from your graph in (b). [Only text is needed for this question]

d. Suppose that we don't like the default order of the categories of `direction_somerville`. It can be helpful to store categorical variables using the **factor** data type - this allows us to set the possible categories and their order explicitly.

Run the code below to manually reorder the categories by turning `direction_somerville` into a factor (as well as remove observations that have missing values of `direction_somerville`). What do you think the default ordering of categories was? Recreate the bar graph of proportions from (b) with this new ordering for `direction_somerville` in the chunk below. [Text and code are needed for this question]

```
# Remove missing values of `direction_somerville`
somerville <- somerville %>%
  filter(!is.na(direction_somerville))

# Reorder the categories of `direction_somerville`
somerville$direction_somerville <- factor(somerville$direction_somerville,
                                          levels = c("Wrong track",
                                                    "Not sure",
                                                    "Right direction"))

# Recreate the plot below
```

Exercise 2: Visualizing somerville with scatterplots

a. Create a scatterplot that shows the relationship between `satisfied_somerville` and `convenient_transport`, and where color is determined by `direction_somerville`. On the *x*-axis, put the variable that you think may *explain* the behavior of the other variable. [Only code is needed for this question]

b. It is difficult to detect a meaningful pattern in the scatterplot from part (a); since the responses were given as whole numbers on a 1 - 10 scale, many of the responses are overlaid on top of each other. Recreate the plot from part (a) by using a new geom, `geom_jitter()`, rather than the `geom` used previously and by making the `geom` somewhat transparent to avoid overplotting. [Only code is needed for this question]

Exercise 3: Airline Delay

You're shopping for a plane ticket and are deciding between three airlines that each have a flight to your destination and are all scheduled to leave at the same time and arrive at the same time. Below are the delays in arrival times (in minutes) for 10 flights from each of the three airlines (a negative number means the flight arrived early, 0 means the flight arrived exactly on time, a positive number means the flight arrived late):

```
airline1 <- c(-60, -45, -30, -15, 0, 0, 15, 30, 45, 60)
airline2 <- c(0, 0, 5, 5, 5, 5, 5, 5, 10, 10)
airline3 <- c(-10, -10, -10, -10, -10, -10, -10, -10, -10, 90)
```

a. Describe each airline's delay record by calculating summary statistics in R (recall the functions `mean()`, `var()`, `sd()`, `median()` from lecture). What each airline's strengths? What are their weaknesses?

b. Which airline would you choose? What characteristics of the airline most factored into your decision?

c. In general, what is the difference between the mean and the median? When should we expect them to be different?
